

Special topics in logic and security II *

May 24, 2024

1. The code C consists of the words (w, x, y, z) of length 4 over the alphabet $\mathbb{F}_2$ that satisfy the system:

$$\begin{cases} w = y \\ x = z \end{cases}$$

- (a) Make a list of the code-words.
(b) How many words of every possible weight do really occur in the code? Write down the weight polynomial.
2. Let C be the code from the previous exercise and let $C^\perp$ be its orthogonal complement.
(a) Apply the Theorem of MacWilliams in order to find the weight polynomial of $C^\perp$.
(b) List the code-words of $C^\perp$ and verify that the coefficients of the weight polynomial really correspond with the number of words of every given weight.
3. For $U : \mathbb{C}^n \rightarrow \mathbb{C}^n$ linear, let $U^* = \overline{U}^T$.
(a) Let $U : \mathbb{C}^n \rightarrow \mathbb{C}^n$ linear. Show that $\langle Ux|y \rangle = \langle x|U^*y \rangle$ for all $x, y \in \mathbb{C}^n$.
(b) Recall that a unitary transformation is a linear transformation that satisfies $U^* = U^{-1}$. Show that a unitary transformation preserves angles and norms.

4. Show that the following matrix is unitary:

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

5. Is the following matrix unitary?

$$S_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

6. Compute:

$$\begin{pmatrix} I_2 & O & O & O \\ O & I_2 & O & O \\ O & O & I_2 & O \\ O & O & O & M_- \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

This operation has an interpretation in mathematical logic. Explain this interpretation.

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